Do Now · retrieval

Lesson 068 · Backtracking: Choices, Dead Ends and Undo

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Describe the divide-and-conquer approach to solving a problem.

Q2

What is a dead end in a search?

Q3

Why must recursion have a base case?

Q4

Stack Push 2, Push 5, Pop, then Push 8: give popped item and remaining stack.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Define and predict — Backtracking explores a choice, tests whether its state remains viable, then undoes the choice if it cannot lead to a solution.

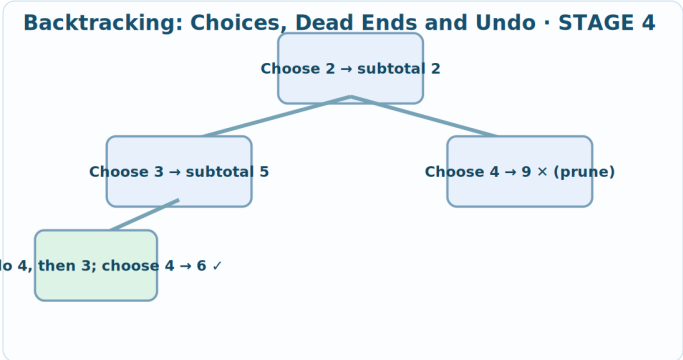
Core idea 2

Explain — A search tree records choices and alternative branches. Pruning avoids expanding branches already proven invalid.

Core idea 3

Evaluate — A correct search needs a stopping condition. Pruning on sum>target only works when remaining numbers cannot reduce the sum (e.g. non-negative inputs).

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

backtracking — Try and undo choices.

search tree — Branches of possible decisions.

pruning — Stop provably impossible branch.

state — Current partial solution data.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

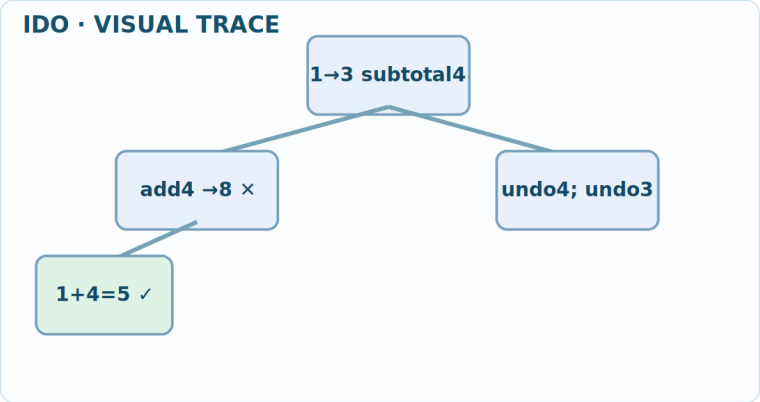
Problem

Find a subset of [1,3,4] with total 5.

Follow the worked method

1. Follow the reasoning — Return from the failed choice of4 before excluding3. The state returns to [1], then choosing4 reaches5. Recursive local totals restore on return; mutable state needs an explicit undo.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Follow the reasoning
The alternative branch 2+7 gives a valid solution.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace first failed and one successful branch for subset sum target6 from [2,3,4]. [6]

Question 2

Explain a recursive backtracking design with choose/exclude and two stopping conditions. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why pruning when sum>target may be invalid if negative integers occur; give counterexample. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A branch is always impossible when its subtotal exceeds the target, even if negatives remain.

Use key vocabulary from memory

1. Explain backtracking in your own words.

2. Explain search tree in your own words.

3. Explain pruning in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.